Thiago Zilli

Data Scientist · Analytics & BI
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Data Scientist with experience in commercial analytics and CRM in manufacturing. At RCM Cabos Elétricos, I built from scratch a daily, auditable data pipeline, 3 executive dashboards and a FastAPI service, replacing scattered Excel spreadsheets with a single source of truth. Developed a monthly revenue forecasting model with Prophet (~9% mean error, MAPE). B.Sc. in Physics from UFPR, where I led a Top 3 team at the university's 1st Data Hackathon. Visiting researcher at the University of Lethbridge (Canada) and first author of a paper on computer vision and fine-tuning of a foundation image-segmentation model.

Experience

Data Scientist (Freelance) — RCM Cabos Elétricos Jan 2025 – Aug 2026
Remote, Curitiba, Brazil

- Replaced scattered spreadsheets, whose KPIs disagreed across reports, with a single source of truth for leadership and sales teams.
- Interviewed two departments to translate business pain points into 9 objectives, and cut 3 unfeasible ones with documented justification before writing any code.
- Single-handedly built from scratch a daily, auditable pipeline over two PostgreSQL databases (logs and import reports); cut its memory usage by 80%, eliminating crashes on 8 GB machines.
- Built from scratch 3 Streamlit dashboards (company overview, sales reps and clients) tracking revenue, profit, margin, conversion and quotas, with period-over-period comparison and drill-down to client, product, state and sales rep.
- Developed LRFM segmentation, cross-selling and LLM-based sentiment classification to flag at-risk clients and sales opportunities.
- Created a statistical detector for discounts above approval limits, turning a leadership suspicion into quantified, auditable evidence.
- Developed a monthly revenue forecast with Prophet (~9% MAPE, up to 3 months ahead).
- Built from scratch, in 2 weeks, a FastAPI REST API with 203 endpoints, domain-modular architecture, Pydantic v2 validation and Swagger docs, unlocking analytics previously trapped in Streamlit; a 704-call smoke test (~55 s) caught and fixed 2 real bugs.

Visiting Researcher (Mitacs Globalink Research Award) — University of Lethbridge Aug 2026 – Nov 2026
On-site, Lethbridge, Canada

- Research on complex-valued oscillatory neural networks, studying how transient network dynamics perform computation, supervised by Prof. Roberto Budzinski.
- Carried out an ***AI-driven transpilation*** project, translating, documenting and testing an entire MATLAB codebase into Python, creating a learning tool for all lab members.

Project Manager — Mathmorphosis Junior Enterprise (UFPR) Jul 2025 – Dec 2025

- Led a 3-person team at UFPR's 1st Data Hackathon: in 5 days, we built a visualization platform on real institutional-evaluation data, with a sentiment index and opinion-balance charts. Top 3 out of 10+ teams; the solution was considered for future use by the university.

Research & Publication

Fine-tuning Cellpose-SAM for neuron segmentation and morphometry — UFPR, with the Pharmacology department (first author; under review at *Neuroinformatics*; presented at the International Headache Congress 2025)

- With only 31 annotated images, raised AP@50 (average precision) from 0.129 to 0.747 and reached an F1-score of 0.85 for neuron counting, using k-fold validation, data augmentation and grid search.
- Cut image segmentation time from ~15 minutes to under 10 seconds per image, with quantified error on morphometric measurements.

Undergraduate Research — UFPR (Nov 2023 – Apr 2026): molecular dynamics of pentagraphene and synthesis/characterization of ZnO nanoparticles (NAPI Zero-Carbon Energy; poster at B-MRS 2025).

Education

B.Sc. in Physics — Federal University of Paraná (UFPR), Brazil 2022 – Dec 2026

Relevant coursework: Programming; Introduction to Statistics; Algorithms and Data Structures I; Computational Physics Methods. Non-degree graduate student in Deep Learning (UFPR).

Academic Mobility (Internship Program) — University of Lethbridge, Canada 2026

Highlighted Courses

- **Data Science Track** — Alura (46h): exploratory analysis, visualization, hypothesis testing, linear regression and time series.
- **SQL with PostgreSQL** — Alura (28h): DML/DDL, views, subqueries, functions and data analysis with SQL in Python.
- **Streamlit: building an interactive dashboard** — Alura (10h)
- **Time Series and Prophet** — Alura (16h): time series analysis and forecasting with Prophet.
- **Classification with Machine Learning** — Alura (16h): models, validation and evaluation metrics.

Supplementary: Python OOP and APIs Track (22h); Regression with variable transformation (8h); Keras: neural networks (8h); NLP sentiment analysis (6h) — Alura · Python & Databases — LIneA (10h) · Data Science Summit 2024 (18h).

Technical Skills

- **Data Analysis & BI:** Python (pandas, NumPy), SQL (PostgreSQL, SQLAlchemy), Streamlit, Plotly, Matplotlib; drill-down dashboards, sales KPIs (revenue, margin, conversion, quotas), customer segmentation (LRFM), churn reduction, data storytelling
- **Statistics, ML & NLP:** Prophet (time series), scikit-learn, regression, classification, hypothesis testing, cross-validation (k-fold), grid search; LLM-based sentiment analysis (OpenAI API); PyTorch and foundation-model fine-tuning
- **Data Engineering:** FastAPI, Pydantic v2, Swagger, Docker / Docker Compose, scheduled pipelines with logging and auditing, automated testing, Git
- **Methodology:** CRISP-DM, MVC, Clean Code / SOLID
- **Languages:** Portuguese (native); English (advanced, professional use)